

Jaya Kiran Kurade

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Master's candidate in Computer Science with 2 years of hands-on experience in AI/ML research and software development. Proficient in building innovative, data-driven, and scalable solutions. Actively seeking opportunities in AI/ML engineering and software development to drive impactful real-world technologies.

EDUCATION

M. S. Computer Science

California State Polytechnic University, Pomona, CA, USA

Expected May 2026

Current GPA 4.0

B. E. Computer Science & Engineering

Visvesvaraya Technological University, Belgaum, India

May 2022

GPA 3.7

Courses: Computer Vision, Connected and Autonomous Vehicles, Advanced Algorithm Design and Analysis, Scheduling Algorithms, Human Computer Interaction, Advanced Software Engineering, Artificial Intelligence & Machine Learning, Complex Analysis, Probability & Statistical Methods

TECHNICAL SKILLS

AI & ML: CNN, DWT, Optical Character Recognition, TensorFlow, PyTorch, Keras, OpenCV, Pandas, NumPy, Jupyter, MATLAB, NLP

Languages & Technologies: Python, PowerShell, C, C++, HTML, CSS, SQL, AWS Cloud, GitHub, Git, Docker, GIS

Databases & Frameworks: MySQL, PostgreSQL, Django, React JS, Streamlit

Other: ServiceNow Platform, Windows, Linux & Network Administration, VMware & Citrix Virtualization, Analytical Skills

PROFESSIONAL EXPERIENCE

Graduate Research Assistant at Cal Poly Pomona, CA, USA

Feb 2025 - Present

Funded by Agriculture Research Institute (ARI), California Govt.

Conducting research on AI/ML-based image fusion to integrate UAV and Satellite imagery under expert mentorship in an ARI-funded initiative, focusing on generating high-resolution imagery for precision agriculture, crop monitoring, and sustainable farming in California.

Technologies: CNN, DWT, ArcGIS Pro, NDVI, Python, ML

- Using AI/ML, including CNN and DWT, to enhance image clarity and support precision farming, with analysis in ArcGIS Pro and calculation of NDVI values.
- Fusing UAV and Satellite imagery to generate high-resolution agricultural maps, enhancing crop monitoring, and promoting sustainable farming practices in California.

Software Engineer, Capgemini, India

Oct 2022 - Jul 2024

I was responsible for developing and deploying automation bots to reduce manual errors, streamline workflows, and improve efficiency in repetitive tasks.

Technologies: PowerShell, Python, VSCode, GitHub, Putty, SSH

- Developed a storage clearance bot to compress files older than 14 days and free storage beyond 90% capacity within set rules.
- Developed a ServiceNow bot to filter tickets by description and priority, automating employee assignments, reducing errors by 82- 85% and increasing efficiency by 79%.
- Contributed to the development of a bot to automate the full database patching lifecycle, including scheduling and compatibility checks, aimed at reducing manual effort.
- Updated and redeployed existing bots with enhanced code and functionality on the customer server, while also gaining experience in documenting these developments.

ACADEMIC PROJECTS

S3 Audio Transcription & Translation Application

Jun 2025 - Jul 2025

Technologies: Python, Streamlit, AWS S3, Docker, MySQL, GitHub Actions, CI/CD

Developed a web application that gave a solution to improve comprehension and accessibility by providing real-time transcripts and translations for better learning and communication.

- Developed web app with OpenAI Whisper for transcription & multilingual translation; integrated AWS S3 and MySQL.
- Containerized with Docker; implemented CI/CD using GitHub Actions for automated build, test, and deployment.
- Automated AWS EC2 deployment via appleboy/ssh-action; added Pytest unit tests for build reliability.

Text Recognition and Extraction from Images

Jan 2022 - Jul 2022

Technologies: Python, Deep Learning- Keras, TensorFlow, Machine Learning- OpenCV

An efficient neural network for accurately predicting text characters or digits to enhance recognition capabilities in text processing applications.

- Applied expertise in Optical Character Recognition (OCR), Tesseract, and Neural Networks, with a focus on Convolutional Neural Networks (CNNs).
- Developed automated workflows to streamline data extraction processes and improve efficiency.
- Achieved 87.10% accuracy in image-based text extraction, optimizing both accuracy and processing efficiency.

Machine Learning Intern, VI Solutions, India

Aug 2021 - Sep 2021

- Completed a month-long internship in IoT, machine learning, and deep learning.
- Focused on areas like process automation, data acquisition, and visual inspection.

Hostel Management System Database

Dec 2020 - May 2021

Technologies: MySQL, Java, NetBeans

A software solution for hostel management to reduce human errors, ensure data consistency, and minimize manual labor in managing various activities.

- Addressed challenges by mitigating human errors, ensuring data consistency, and minimizing manual labor.
- Facilitated easy access to information for users and staff.